Smart Wallet – AI Based Automatic Expense Categorization System

PROJECT
APRIL
2026

PROJECT REPORT SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE AWARD OF THE DEGREE OF **BACHELOR OF ENGINEERING** IN **COMPUTER SCIENCE AND ENGINEERING** OF THE ANNA UNIVERSITY

PROJECT WORK

Submitted by

GURUVEL P- 722823104305

BATCH
2023 - 2027

Under the Guidance of

Dr. J. SREEMATHY M.E., Ph.D

Assistant Professor, CSE.

COMPUTER SCIENCE AND ENGINEERING

Sri Eshwar College of Engineering

(An Autonomous Institution – Affiliated to Anna University)

COIMBATORE – 641 202

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BONAFIDE CERTIFICATE

Certified that this Report titled **"Smart Wallet – AI Based Automatic Expense Categorization System"** for the course **U23CS493 - OBJECT ORIENTED ANALYSIS AND DESIGN** is the bonafide work of

GURUVEL P **722823104305**

who carried out the project work under my supervision.

SIGNATURE

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Submitted for the **Autonomous Semester End Project Viva-Voce** held on

……..

INTERNAL EXAMINER **EXTERNAL EXAMINER**

DECLARATION

GURUVEL P [722823104305]

To declare that the project entitled **"Smart Wallet – AI Based Automatic Expense Categorization System"** submitted in partial fulfilment to the University as the project work of Bachelor of Engineering (Computer Science and Engineering) Degree, is a record of original work done by us under the supervision and guidance of Dr. J. Sreemathy M.E.,Ph.D, Assistant Professor, Department of Computer Science and Engineering, Sri Eshwar College of Engineering, Coimbatore.

Place: Coimbatore
Date:

GURUVEL P

Project Guided by,
Dr. J. Sreemathy, M.E., Ph.D
Assistant Professor

ACKNOWLEDGEMENT

The success of a work depends on a team and cooperation. I take this opportunity to express my gratitude and thanks to everyone who helped me in my project. I would like to thank the management for the constant support provided by them to complete this project.

It is indeed our great honor bounded duty to thank our beloved **Chairman Mr.R. Mohanram,** for his academic interest shown towards the students.

We are indebted to our **Director Mr. R. Rajaram,** for motivating and providing us with all facilities.

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We are indebted to **Dr. R. Subha, M.E., Ph.D.** Head of Computer Science and Engineering Department for having permitted us to carry out this project and giving the complete freedom to utilize the resources of the department.

I express my sincere thanks to my mini project Coordinator **Dr. J. Sreemathy, M.E.,Ph.D**, Assistant Professors of Computer Science and Engineering Department for the valuable guidance and encouragement given, to us for this project. I solemnly express our thanks to all the teaching and nonteaching staff of the Computer Science and Engineering Department, family and friends for their valuable support which inspired us to work on this project.

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ABSTRACT

The Smart Wallet System is a modern digital solution designed to simplify financial tracking and expense management. It enables users to store, manage, and categorize transactions efficiently through an intuitive interface. Users can monitor spending habits, categorize expenses, and receive real-time insights into their financial activities.

The system is built using a full-stack architecture with a responsive frontend, a robust backend, and a scalable database. It ensures secure data handling, real-time updates, and smooth performance. Additional features such as transaction history, analytics, and notifications enhance user experience.

This project demonstrates how modern technologies can be used to solve real-world financial management problems and improve user awareness of spending behavior.

1.1. Project Overview & Problem Statement

Managing personal finances manually is inefficient and error-prone. Many users struggle to track expenses, categorize spending, and analyze financial habits.

The Smart Wallet system addresses this problem by providing a centralized platform where users can:

- Record transactions
- Categorize expenses
- Monitor spending patterns
- Receive insights and alerts

This system replaces traditional methods like notebooks or spreadsheets with an automated, intelligent solution.

1.2. Objectives

1. **Centralized Financial Platform:** Provide a single platform to manage all income and expense transactions efficiently.
2. **Smart Expense Categorization:** Automatically classify transactions into categories like Food, Travel, Bills, and Shopping.
3. **User-Friendly Experience:** Design an intuitive and responsive interface using modern frontend technologies.
4. **Real-Time Financial Insights:** Enable users to track spending patterns and monitor their financial status instantly.
5. **Secure and Scalable System:** Ensure strong data protection, authentication, and scalability for handling large user data.

2.1. Functional Requirements

1. **Secure Authentication:** User Signup/Login with encrypted passwords and session handling.
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3. **Expense Categorization:** Automatically or manually assign categories to transactions.
4. **Transaction History:** View and filter transactions based on date, category, and type.
5. **Analytics Dashboard:** Display spending insights using charts and reports.
6. **Budget Alerts:** Notify users when spending exceeds predefined limits.

2.2. Non Functional Requirements

1. **High Performance:** Fast loading and quick response time for all operations.
2. **Scalability:** Ability to handle increasing users and transaction data efficiently.
3. **Data Security:** Encrypted passwords and secure APIs for safe data handling.
4. **Responsiveness:** Fully adaptive UI for mobile, tablet, and desktop devices.

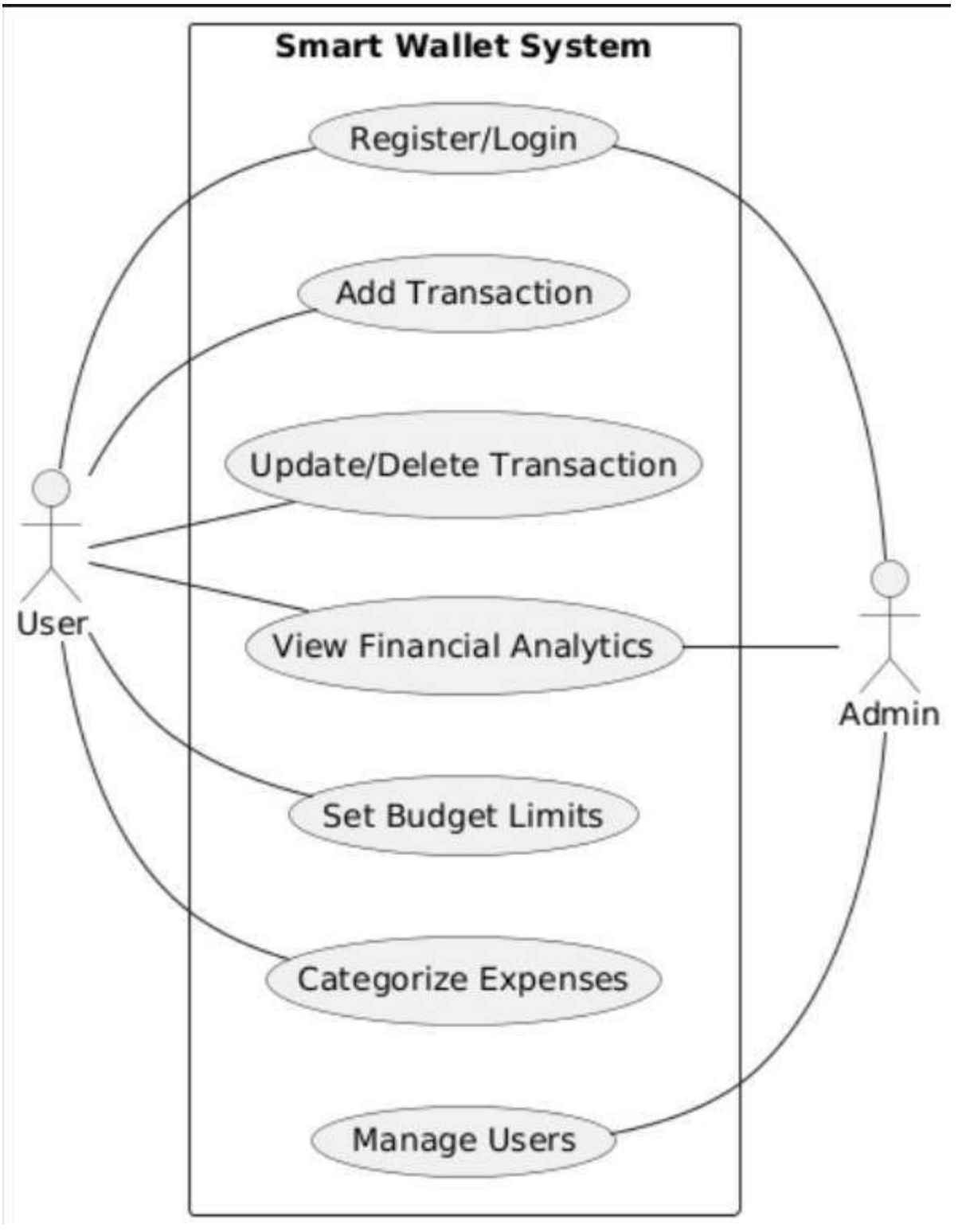
3.1. Tools

1. Frontend
   - **React.js / Flutter:** For building responsive and interactive user interfaces.
   - **HTML5 & CSS3:** For structuring and styling the application.
   - **JavaScript:** For dynamic functionality and user interaction.
2. Backend
   - **Node.js (Express):** Lightweight and fast API handling.
   - **Java (Spring Boot):** Secure and scalable backend processing.
   - **BCrypt:** Password hashing for secure authentication.
3. Database
   - **MongoDB:** NoSQL database for flexible and scalable data storage.**Firebase (Optional):** Real-time database and authentication support.
4. Development Tools
   - IntelliJ IDEA: IDEs for coding frontend and backend.
   - Postman: To test and debug API endpoints.
   - Git & GitHub: For version control and collaboration.

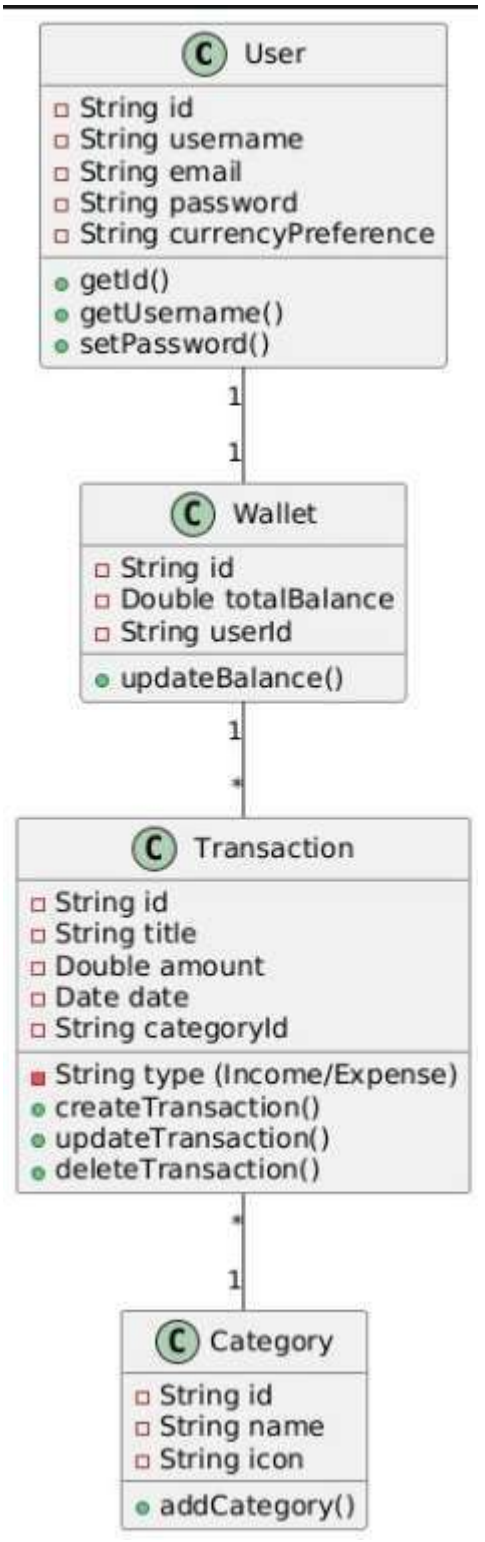
4.1. Design Patterns Used

1. **Builder Pattern**
   - Used to construct transaction objects step-by-step.
   - Improves readability and flexibility in object creation.
2. **Factory Pattern**
   - Used to create different transaction categories dynamically.
   - Simplifies category management.
3. **Observer Pattern**
   - Used for real-time updates like notifications and dashboard refresh.
4. **MVC Pattern**
   - Separates application into Model, View, and Controller.
   - Improves maintainability and scalability.
5. **Composition Pattern**
   - Used for database connections and API instances.
   - Ensures efficient resource usage.

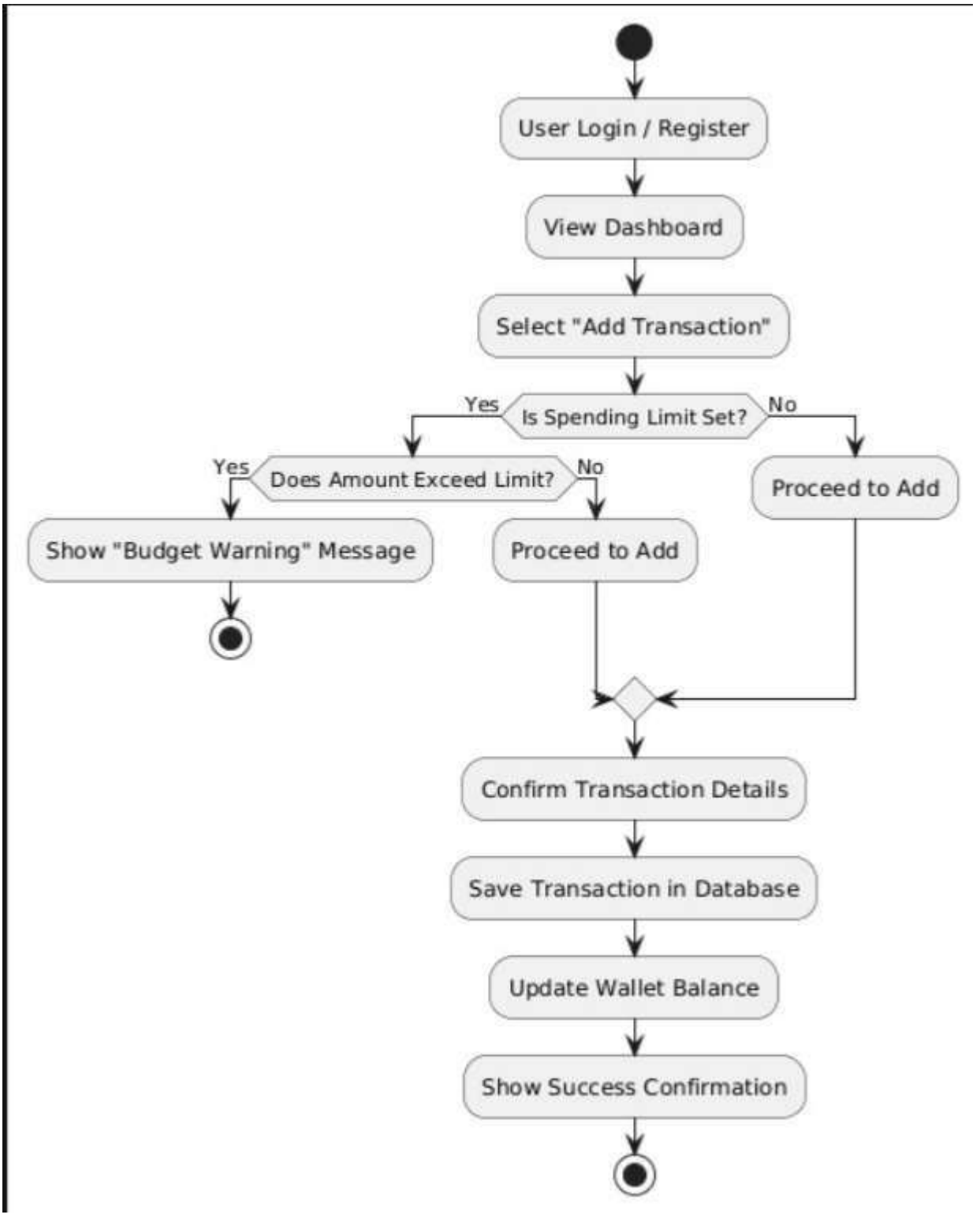
5.1. Usecase Diagram



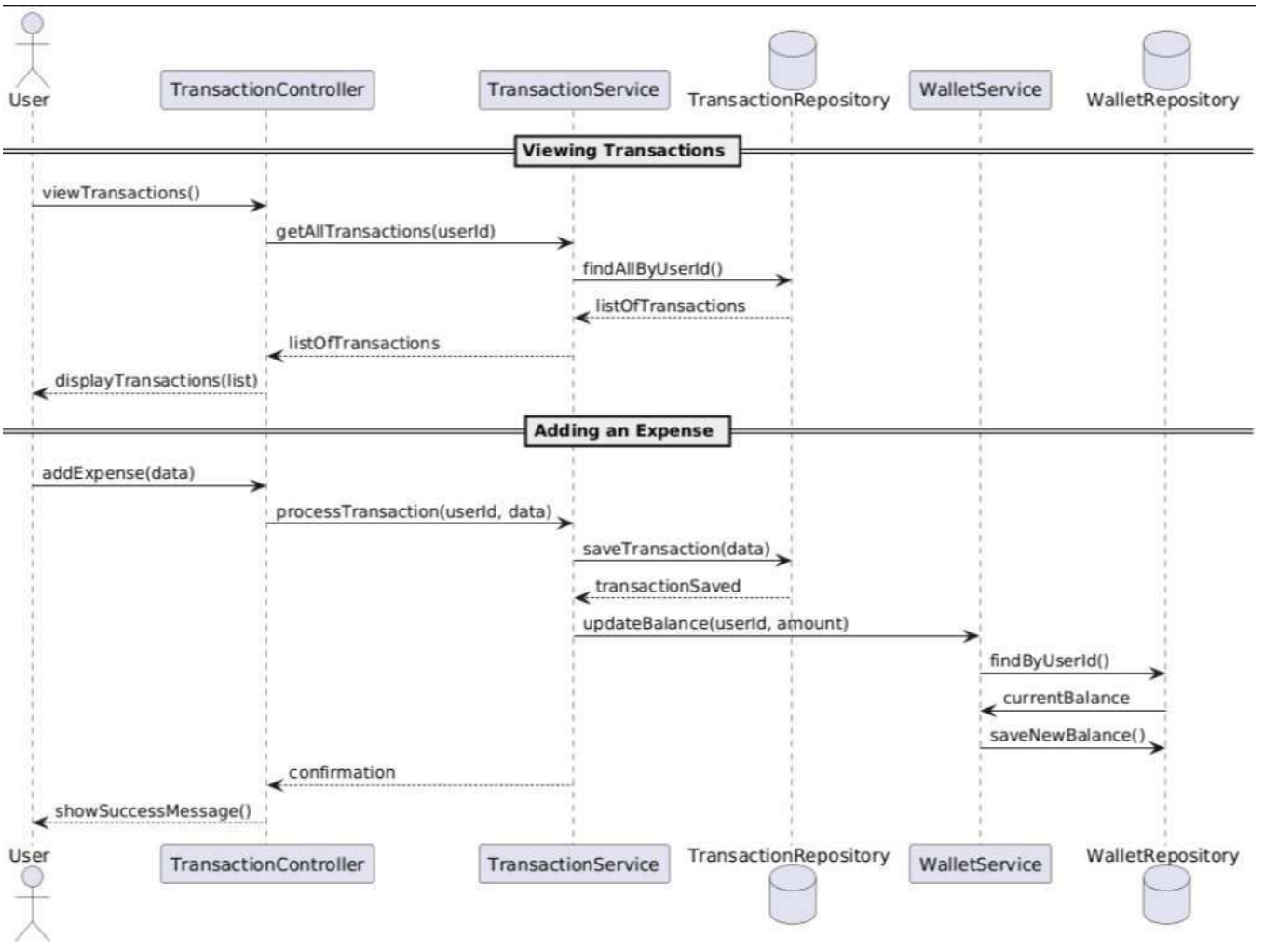
5.2 Class Diagram



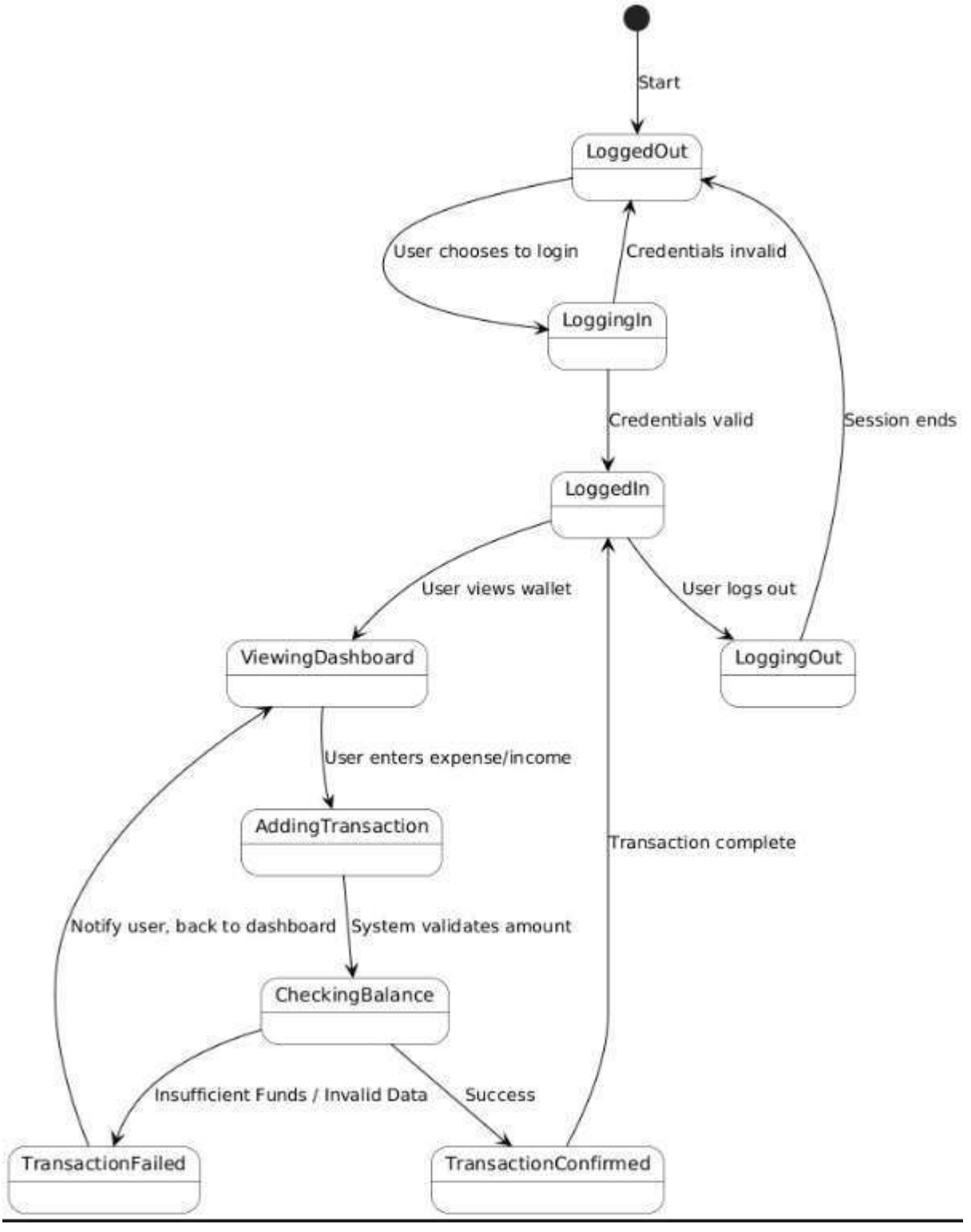
5.3 Activity Diagram



5.4 Sequence Diagram



5.5 State Diagram



6.1. Testing Types Used

- **Unit Testing:**Testing individual components like login and transaction modules.
- **Integration Testing:** Verifying interaction between frontend, backend, and database.
- **System Testing:** Testing the entire workflow of the application.

6.2. Tools Used

- **Manual Testing –** Basic functional checks for UI and workflows.
- **Browser DevTools** – Inspect network calls, console logs, and responsiveness.
- **Selenium WebDriver –** Automated end-to-end UI testing.
- **Postman –** Backend API testing for event and registration endpoints.

6.3. Sample Test Cases

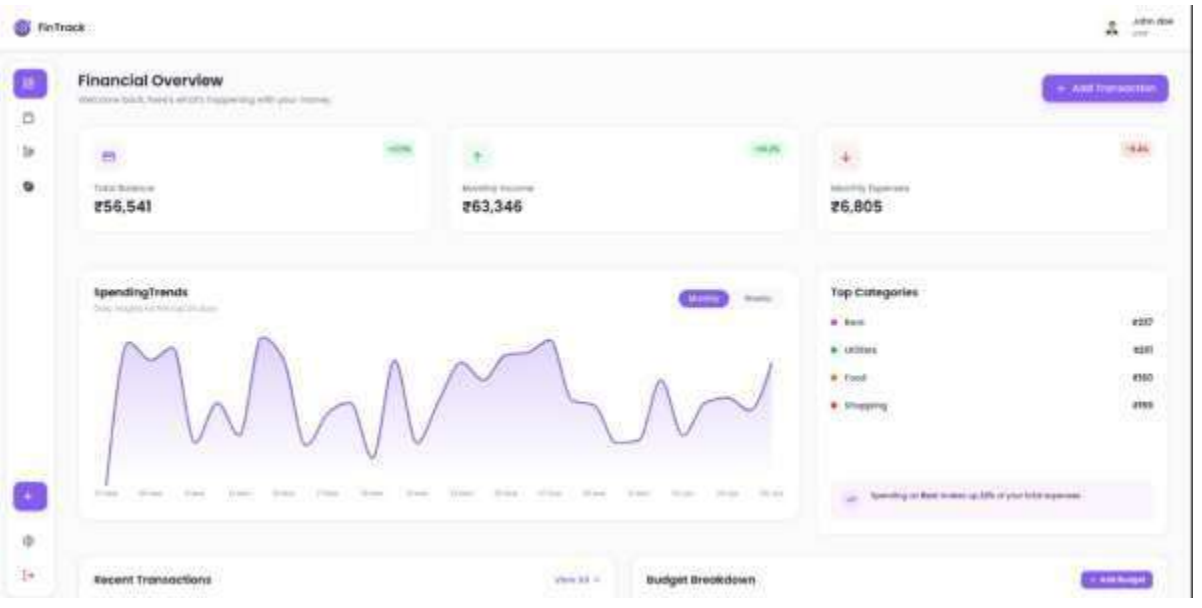
Test Case ID	Module	Description	Expected Result	Status
TC01	Auth	Login with valid credentials	Successful login	Pass
TC02	Transaction	Add expense	Stored correctly	Pass

Test Case ID	Module	Description	Expected Result	Status
TC03	Category	Auto categorization	Correct category assigned	Pass
TC04	Dashboard	Generate report	Accurate analytics	Pass
TC05	Notification	Budget alert	Alert triggered	Pass

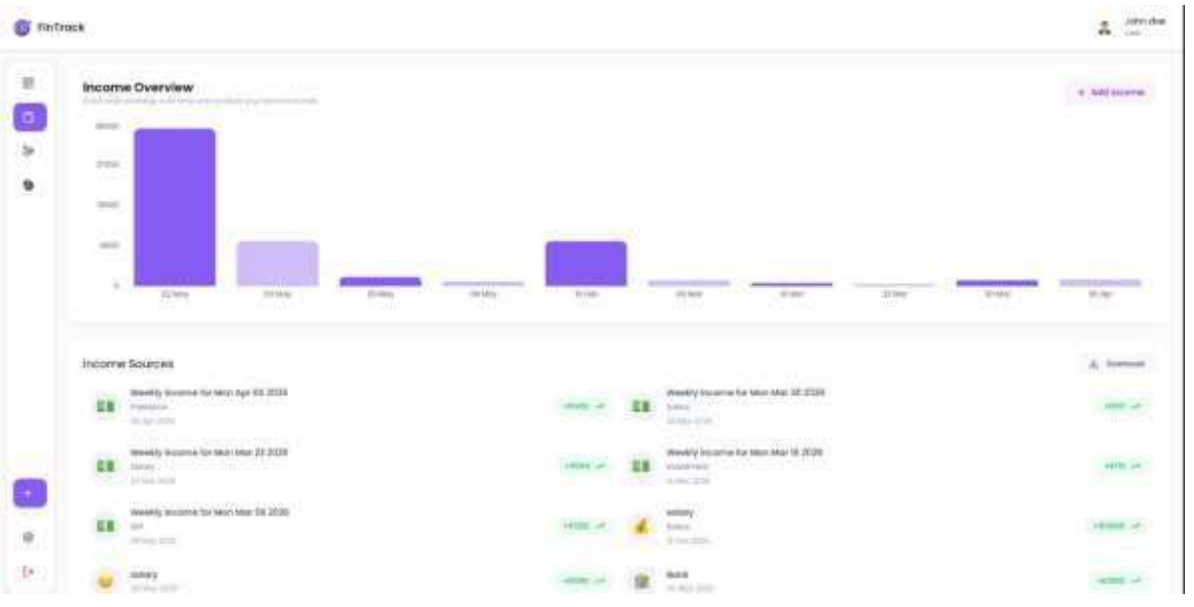
6.4. Test Reports Summary

- All core modules (Login, Transactions, Dashboard, Notifications) passed testing.
- No critical bugs found in the final version.
- Backend APIs returned correct responses consistently.
- UI tested across devices and browsers successfully.

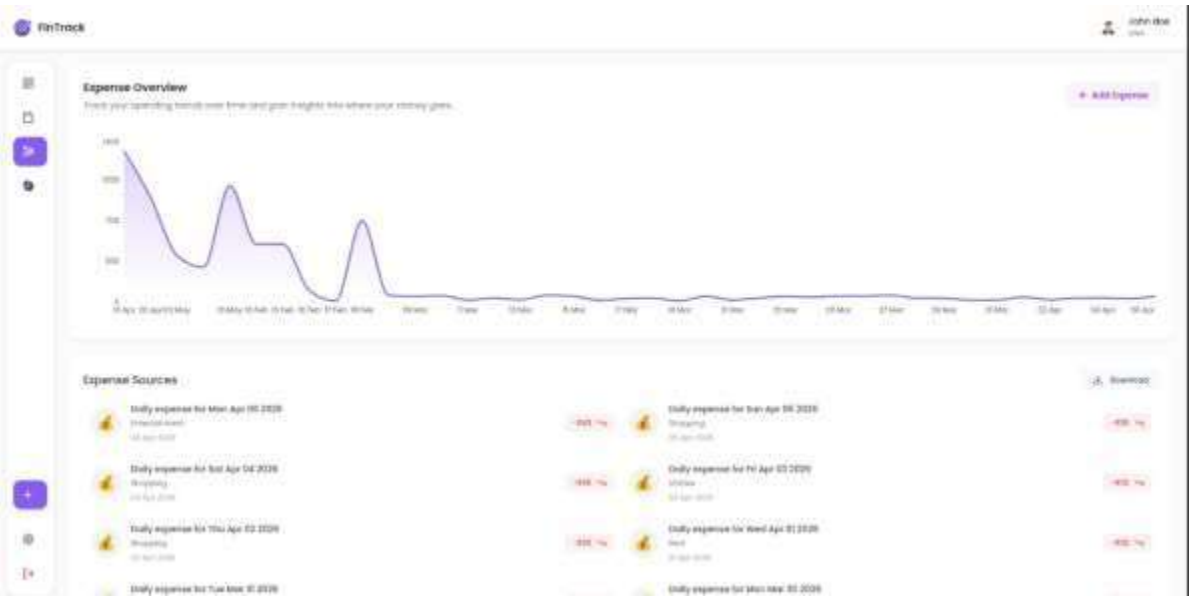
7.1. Home Page



7.2. Income Overview Page



7.3. Expense Dashboard



CONCLUSION

The Smart Wallet project successfully demonstrates an efficient and scalable financial management system. It simplifies tracking of income and expenses while providing meaningful insights through analytics. The system ensures security, performance, and usability through modern technologies and design patterns.

This project highlights practical implementation of full-stack development concepts and provides a strong foundation for future enhancements such as AI-based expense prediction and automated budgeting.

[1] I. Sommerville, *Software Engineering*, 10th ed. Pearson, 2015.

[2] E. Freeman and E. Robson, *Head First Design Patterns*. O'Reilly Media, 2004.

[3] Spring Boot, "Spring Boot Reference Documentation," [Online]. Available: https://spring.io/projects/spring-boot. [Accessed: Apr. 03, 2026].

[4] React, "React – A JavaScript library for building user interfaces," [Online]. Available: https://reactjs.org/docs/getting-started.html. [Accessed: Apr. 03, 2026].

[5] Selenium, "Selenium WebDriver Documentation," [Online]. Available: https://www.selenium.dev/documentation/. [Accessed: Apr. 03, 2026].